

ASSIGNMENT I - BASED ON SDG GOALS

Course Code: ITA0609

Course Title: Machine Learning

Student Name: S.Swagat

1) . Explain traffic congestion prediction using Machine Learning.

Traffic Congestion Prediction Using Machine Learning

SDG 11 – Sustainable Cities and Communities

1. Problem Understanding

Traffic congestion is one of the biggest challenges faced by modern cities. As the number of vehicles on the road increases, traffic jams become more frequent, leading to longer travel times, fuel wastage, increased air pollution, and higher stress levels for commuters. Efficient traffic management is essential for improving the quality of urban life.

Machine Learning (ML) provides an intelligent solution by analyzing historical and real-time traffic data to predict congestion before it occurs. These predictions enable traffic authorities to optimize traffic signals, suggest alternative routes, and improve overall transportation efficiency. The primary objective of this project is to build a Machine Learning model that accurately predicts traffic congestion based on factors such as vehicle count, average speed, weather conditions, time of the day, and road conditions.

2. Data Collection and Preparation

The first step in the project is collecting traffic-related data from reliable sources. Data can be obtained from traffic surveillance cameras, road sensors, GPS-enabled vehicles, Google Maps APIs, government transportation departments, or publicly available traffic datasets.

Input Parameters

- Vehicle Count
- Average Vehicle Speed
- Time of Day

- Day of the Week
- Weather Conditions
- Road Type
- Traffic Signal Status
- Accident Information (if available)

After collecting the data, preprocessing is performed to improve data quality.

Data Preparation Steps

- Remove duplicate records.
- Handle missing values using suitable techniques.
- Convert categorical values (such as weather conditions) into numerical values.
- Normalize or standardize numerical features.
- Split the dataset into training and testing datasets.

A clean and well-prepared dataset helps improve the accuracy and performance of the Machine Learning model.

3. Analysis Using NumPy

NumPy is used for performing numerical computations efficiently. It helps process large traffic datasets and calculate statistical measures required for analysis.

Operations Performed Using NumPy

- Calculate average vehicle count.
- Find maximum and minimum traffic volume.
- Compute average vehicle speed.
- Calculate standard deviation and variance.
- Identify peak traffic hours.
- Perform matrix operations for Machine Learning preprocessing.

After preprocessing and analysis, the dataset is provided as input to Machine Learning algorithms such as:

- Random Forest
- Decision Tree
- Linear Regression
- Support Vector Machine (SVM)
- Neural Networks

The selected algorithm learns traffic patterns from historical data and predicts future congestion levels.

4. Visualization and Interpretation

Data visualization helps understand traffic behavior and identify congestion patterns.

Graphs Used

- Line Graph – Traffic Volume vs Time
- Bar Chart – Vehicle Count by Road
- Scatter Plot – Average Speed vs Traffic Density
- Heat Map – Congestion Levels Across Different Areas
- Pie Chart – Distribution of Traffic Conditions

Interpretation

The graphs help identify:

- Peak traffic hours during mornings and evenings.
- Roads with the highest congestion.
- Areas requiring better traffic signal management.
- Effect of weather conditions on traffic flow.
- Daily and weekly traffic trends.

Visualization enables traffic authorities to make informed decisions for better urban traffic management.

5. Innovation and SDG Mapping

Innovation

This project introduces an intelligent traffic congestion prediction system that uses Machine Learning to provide accurate congestion forecasts. The system can:

- Predict traffic congestion in advance.
- Suggest alternative routes to drivers.
- Optimize traffic signal timings.
- Provide real-time congestion alerts.
- Support integration with Smart City infrastructure.
- Reduce travel time and improve transportation efficiency.

SDG 11 – Sustainable Cities and Communities

This project directly contributes to **United Nations Sustainable Development Goal 11**, which focuses on making cities inclusive, safe, resilient, and sustainable.

Contribution to SDG 11

- Reduces traffic congestion.
- Minimizes fuel consumption.
- Lowers greenhouse gas emissions.
- Improves road safety.
- Supports efficient public transportation.
- Promotes smart and sustainable urban mobility.
- Enhances the quality of life for city residents.

By using Machine Learning for intelligent traffic management, cities can become cleaner, greener, and more efficient.

6. Documentation and Conclusion

The complete project documentation includes the problem statement, dataset description, preprocessing methods, NumPy-based analysis, Machine Learning model development, prediction results, and visualization outputs. The system is tested using training and testing datasets to evaluate prediction accuracy.

Conclusion

Traffic Congestion Prediction using Machine Learning is an effective solution for managing urban traffic problems. By analyzing historical and real-time traffic data, the system accurately predicts congestion and supports better traffic management decisions. NumPy enables efficient numerical analysis, while data visualization provides meaningful insights into traffic patterns. The project significantly contributes to **SDG 11 – Sustainable Cities and Communities** by reducing congestion, lowering pollution, improving road safety, and promoting smart transportation systems. This approach helps create sustainable, efficient, and environmentally friendly cities for the future.